BENNETT UNIVERSITY (THE TIMES GROUP), GREATER NOIDA

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND TECHNOLOGY (SCSET)

Project Report on

"Customer Complaint Sentiment Analysis and Category Analysis"

Submitted in AI ML Project Course, CSET 312

Bachelor of Technology

MILESTONE 2 REPORT

Model Implementation, System Architecture, and Preliminary Results

Team Member	Enrollment No.
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1. Executive Summary & Project Overview

Modern financial services, banking institutions, and fintech enterprises handle high daily volumes of customer complaints. Manual inspection and routing of grievances lead to operational bottlenecks, delayed responses to urgent fraud incidents, and reduced customer satisfaction. The primary objective of this project is to build an end-to-end **AI-based Customer Complaint Intelligence System** that automates text preprocessing, multi-class issue categorization, urgency priority estimation, and targeted operational department routing.

During **Milestone 2**, the team successfully transitioned from initial problem formulation to a fully functional, integrated application. The project artifacts include a validated NLP text preprocessing pipeline, a TF-IDF feature extraction engine, a trained Logistic Regression baseline model, an automated rule-assisted priority escalation module, a persistent SQLite database schema, a FastAPI RESTful backend API, and an interactive Next.js 16 web interface supporting role-based portal access and dark/light theme switching.

2. System Architecture & Conceptual Workflow

The end-to-end data and operational flow of the system follows a structured 7-stage architectural pipeline:

- **1. Intake Portal:** Customers submit unstructured natural language complaint narratives via the web client.
- **2. NLP Preprocessing:** Text cleaning, lowercasing, punctuation stripping, and token normalization.
- **3. Vectorization:** Sublinear term frequency TF-IDF encoding converting text into numerical feature vectors.
- **4. AI Categorization:** Trained Logistic Regression classifier predicts the specific intent/category.
- **5. Priority & Department Routing:** Urgency engine calculates priority (High/Medium/Low) based on intent and emergency flags, then assigns the ticket to specialized departments.
- **6. Database Persistence:** Records are stored in a SQLite relational database with real-time status tracking.
- **7. Admin Intelligence Dashboard:** Support teams filter, monitor, update ticket statuses, and analyze priority distributions.

Pipeline Stage	Component	Technical Execution / Output
Intake & API	FastAPI / Next.js	REST endpoint POST /api/complaints accepting JSON payload
NLP Feature Pipeline	Scikit-Learn TF-IDF	Serialized tfidf_vectorizer.pkl (5,000 max features)
Machine Learning	Logistic Regression	Serialized logistic_regression_model.pkl model
Rule Engine	Priority Evaluator	Flags unauthorized charges & stolen cards as HIGH priority
Data Storage	SQLite DB	complaints.db schema storing tickets & audit timestamps

3. Dataset Description & Preprocessing Methodology

Dataset Sources: The project utilizes real-world consumer complaints sourced directly from the *U.S. Consumer Financial Protection Bureau (CFPB)* database (containing over 277,000+ raw records and a 48,990 clean evaluation corpus). The dataset spans major financial product domains and federal regulatory categories:

- **Credit Reporting & Debt:** Credit reporting discrepancies, debt collection harassment, identity theft disputes.
- **Mortgages & Housing Finance:** Escrow misapplication, loan modifications, foreclosure risk escalations.
- **Payment Cards & Accounts:** Fraudulent debits, stolen cards, checking overdrafts, ATM machine failures.

- **Transfers & Lending:** Delayed international SWIFT wires, student loan forgiveness disputes, auto lease repossessions.

Text Preprocessing Pipeline: Raw complaint narratives undergo rigorous text cleaning:

1. *Case Normalization:* Conversion of all text tokens to lowercase.
2. *Noise & Special Character Removal:* Regex stripping of HTML tags, punctuation, special symbols, and extra whitespace.
3. *Stopword Filtering:* Removal of non-informative English stop words to emphasize discriminative terms.
4. *TF-IDF Feature Extraction:* Term Frequency-Inverse Document Frequency scaling with sublinear term frequency enabled, bi-gram range (1,2), and feature dimension capped at 5,000 top terms.

4. Model Implementation & Preliminary Results

For Milestone 2, a multi-class **Logistic Regression model** with L2 regularization (`solver='lbfgs'`, `C=1.0`) was implemented as the core baseline classifier. The model was evaluated using standard multi-class classification metrics on an 80/20 train-test split:

Evaluation Metric	Score / Value	Description & Analytical Notes
Classification Accuracy	83.2%	Multi-class accuracy across 10 major CFPB financial domains
Macro Precision	0.678	Balanced precision reducing false positive department dispatches
Macro Recall	0.635	High recall capturing nuanced, lengthy consumer narratives
Weighted F1-Score	0.827	Strong overall performance on imbalanced real-world distributions
Inference Latency	< 5 ms	Sub-millisecond inference optimal for real-time web intake

Case Study Validation: Sample complaints evaluated by the pipeline produced high-confidence dispatches:

- *Sample 1:* 'Equifax is reporting a delinquent $1,450 collection from an unknown medical provider that does not belong to me.' → Predicted Category: **Credit Reporting & Repair** | Priority: **HIGH** | Department: **Credit Dispute Bureau**.
- *Sample 2:* 'Someone used my credit card without my permission to purchase $850 worth of items online.' → Predicted Category: **Credit & Debit Cards** | Priority: **HIGH** | Department: **Card Security Team**.

5. Milestone 2 Completion Summary & Team Contributions

All planned deliverables for **Milestone 2** have been completed successfully. The summary of completed milestones and specific team contributions is detailed below:

Team Member	Enrollment No.	Milestone 2 Core Responsibilities & Deliverables
Pranjal Chaturvedi	S24CSEU2082	Data preprocessing pipeline implementation, noise stripping regex, TF-IDF vectorizer configuration, multi-class Logistic Regression model training, and priority rule logic integration.
Tanmay Awal	S24CSEU2098	System architecture design, FastAPI REST API development (`backend/main.py`), SQLite DB schema & seeding, Next.js 16 UI frontend implementation, and project report compilation.

6. Future Scope & Milestone 3 Roadmap

Building upon the successful Milestone 2 deployment, the planned roadmap for Milestone 3 includes:

- **Deep Learning Models:** Fine-tuning DistilBERT and BERT-base transformer models to compare semantic representation against the TF-IDF baseline.
- **Enhanced Analytics:** Integrating Recharts visual charts in the Admin Dashboard for category breakdown and real-time SLA metrics.
- **Automated Escalations:** Email notification webhooks for High-Priority fraud complaints and automated resolution feedback loops.